Solution Architecture Template

Date	15 March 2026
Team ID	TM-OPTICROP-04
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Proposed Solution Architecture & Technology Stack (4 Marks)

Use the below template to outline your architectural layers, component structures, and selected technology framework configurations.

Architectural Layer	Proposed Technology Stack	Description/Core Responsibilities
User Interface (Frontend)	HTML5, CSS3, JavaScript (ES6+), Bootstrap, Chart.js	Provides farmers and administrators with diagnostic dashboards, analytical visual representations of crop cycles, soil metric tracking interfaces, and configuration settings.
Application Core (Backend)	Python, Flask / FastAPI, Gunicorn	Handles business logic pipelines, processes system ingestion feeds, coordinates authentication controls, and routes programmatic calls to computational modules.
Machine Learning / Engine Layer	Scikit-Learn, XGBoost, Pandas, NumPy	Executes predictive algorithms, production optimization logic, yield forecasting models, and analytical assessments of agricultural sensor vectors.
Data Architecture (Database)	PostgreSQL, SQLite (Local Development)	Manages relational structures or document indexes capturing historical climatic records, soil profile properties, user telemetry, and telemetry benchmarks.

System Architecture Design Layout

[Place System Architecture Diagram Here / Insert Conceptual Block Architecture Model Diagram]